Daily Algebra Drill #1

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Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

New toolkit today: Difference / Sum of Squares & Cubes · Newton's Identity Chain · Disguised Quadratics · Telescoping & Partial Fractions.

Attempt each question before checking answers. Time yourself. No calculators.

Section A Rapid Recognition (≤15 s each)

Factorise, simplify, or evaluate. No expansion. Pure recognition.

1. Factorise completely: $x^4 - 16$.

2. Simplify: $\dfrac{a^2 - b^2}{a^2 + 2ab + b^2}$.

3. Without expanding, evaluate: $103^2 - 97^2$.

4. Factorise: $2x^2 + 7x - 15$.

5. Simplify: $\left(x + \dfrac{1}{x}\right)^2 - \left(x - \dfrac{1}{x}\right)^2$.

6. Factorise: $x^3 + 8$.

7. Simplify: $\dfrac{x^2 - 5x + 6}{x^2 - 4}$.

8. Given $a + b = 5$ and $ab = 3$, find $a^2 + b^2$.

9. Factorise: $4x^2 - 12x + 9$.

10. Simplify: $\dfrac{1}{1 \cdot 2} + \dfrac{1}{2 \cdot 3} + \dfrac{1}{3 \cdot 4} + \cdots + \dfrac{1}{9 \cdot 10}$.

Section B Manipulation Drills (≤50 s each)

Algebraic fractions, rearranging, hidden structure. Resist the urge to expand.

1. Simplify: $\dfrac{x^3 - 1}{x^2 - 1}$.

2. Simplify: $\dfrac{1}{x - 1} - \dfrac{2}{x^2 - 1}$.

3. Make x the subject: $\dfrac{x+a}{x-a} = k \quad (k \neq 1)$.

4. Given that $a+b+c=0$, express $(a+b)(b+c)(c+a)$ in terms of abc.

5. Simplify $\dfrac{x^2+3x+2}{x^2-x-2}$ and state all values of x for which the original expression is undefined.

6. Simplify: $\left(\dfrac{x^2}{y} - \dfrac{y^2}{x}\right) \div \left(\dfrac{x}{y} + 1 + \dfrac{y}{x}\right)$.

7. Factorise: $a^2(b-c) + b^2(c-a) + c^2(a-b)$.

8. Solve for x: $\dfrac{3}{x+1} + \dfrac{4}{x+2} = \dfrac{11}{(x+1)(x+2)}$.

9. Simplify: $\dfrac{2^{n+2} + 2^n}{2^{n+1} + 2^{n-1}}$.

10. Given $p+q=7$ and $p^3+q^3=133$, find pq.

Section C Substitution & Structure (≤90 s each)

Look for symmetry, disguised quadratics, and useful substitutions before computing.

1. Solve: $x^4 - 5x^2 + 4 = 0$. *(after TMUA 2023 Paper 2 Q18)*

2. Given $x + \dfrac{1}{x} = 3$, find $x^3 + \dfrac{1}{x^3}$.

3. Find all real solutions to: $(x^2-3x)^2 - 2(x^2-3x) - 8 = 0$.

4. Simplify: $\dfrac{x^2+y^2}{xy} - \dfrac{(x-y)^2}{xy}$.

5. Factorise $x^4 + 4y^4$ over the integers.

6. Given $\sqrt{2x+1} + \sqrt{2x-1} = t$ with $t > 0$, find x in terms of t.

7. Given $a+b+c=0$, evaluate $\dfrac{a^2}{bc} + \dfrac{b^2}{ca} + \dfrac{c^2}{ab}$.

8. Solve the system: $x+y=5, \quad x^2+y^2=17$.

Section D Challenge (TMUA / SMC difficulty)

Insight over computation. Each question has a short, elegant solution — find it.

1. Find the value of

$$\frac{(2025)^3 - (2024)^3 - (2023)^3 + (2022)^3}{(2025)^2 - (2022)^2}.$$

2. Let $f(x) = \dfrac{x}{x+1}$. Define f^n as f composed with itself n times. Find a closed form for $f^n(x)$ and hence compute $f^{2025}(1)$.

3. Positive reals a, b, c satisfy $abc = 1$. Simplify:

$$\frac{1}{1+a+ab} + \frac{1}{1+b+bc} + \frac{1}{1+c+ca}.$$

4. Without solving for x, find $x^2 + \dfrac{1}{x^2}$ given $x - \dfrac{1}{x} = \sqrt{5}$. Hence find $x^4 + \dfrac{1}{x^4}$.

5. Find all integers n such that $n^2 + 4n + 3$ is a perfect square.

"Speed comes from recognition, not from rushing. Every shortcut is a pattern you have internalised."

Found a mistake or want to contribute a sheet?
Visit the open-source project at github.com/The-CerealDev/speed-maths